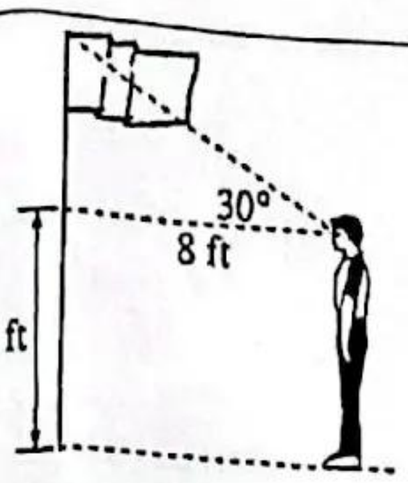
Course Title: Mathematical Analysis for Computer Science **Course Code: CSE-3201**

1. Write down the recursive formula for the Tower of Hanoi and Fabonacci numbers problem.

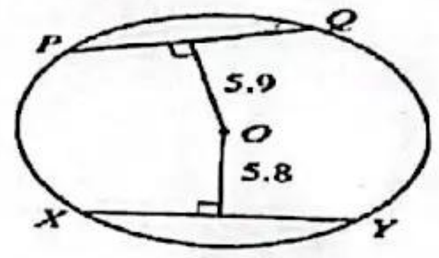
2. For each questions, if you have 2 columns then it represents A and B and your answers means:

A if the quantity in Column A is greater; B if the quantity in Column B is greater; C if the two quantities are equal; D if the relationship cannot be determined from the information given

2. a)
In the figure, a student whose eyes are 6 feet above the ground and 8 feet from a vertical flagpole views the top of the flagpole at a 30-degree angle of elevation. What is the height, in feet, of the flagpole?



b)

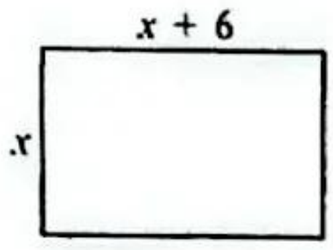


Circle with center O

The length of chord PQ

The length of chord XY

c)

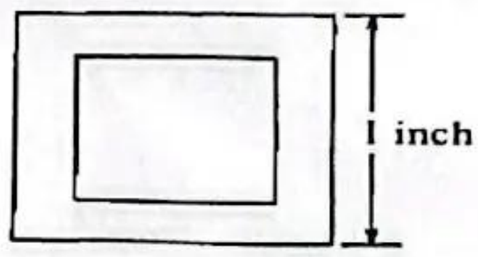


The perimeter of square S is equal to the perimeter of the rectangle above.

The length of a side of S

$x + 3$

d)



In the figure above, if the area of the smaller square region is $\frac{1}{2}$ the area of the larger square region, then the diagonal of the larger square is how many inches longer than the diagonal of the smaller square?

e)

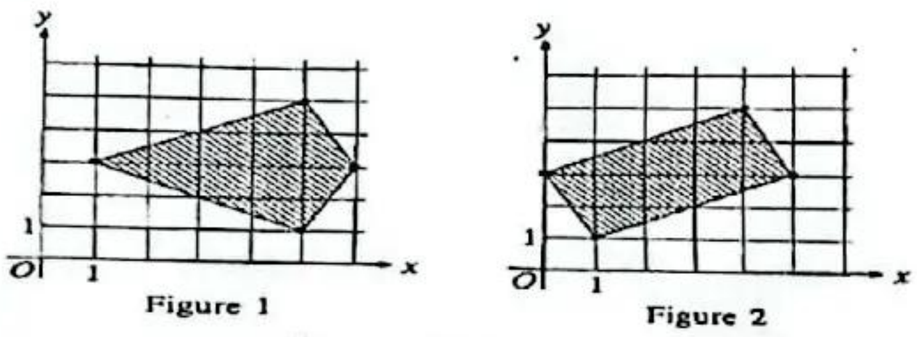


Figure 1

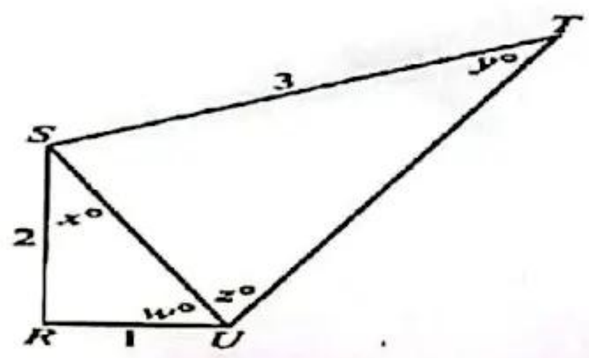
Figure 2

Note: Drawn to scale.

The area of the shaded region shown in Figure 1

The area of the shaded region shown in Figure 2

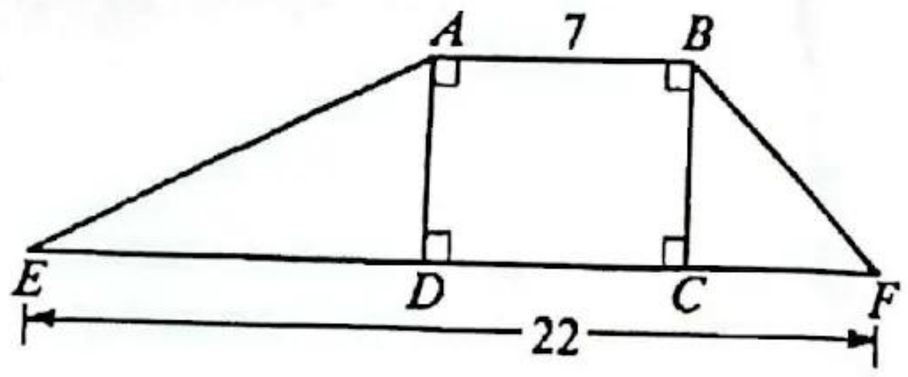
f)



$w + x = 90$
$y + z = 90$

The perimeter of $RSTU$ 10

g)



The area of rectangular region $ABCD$

The area of triangular region ADE

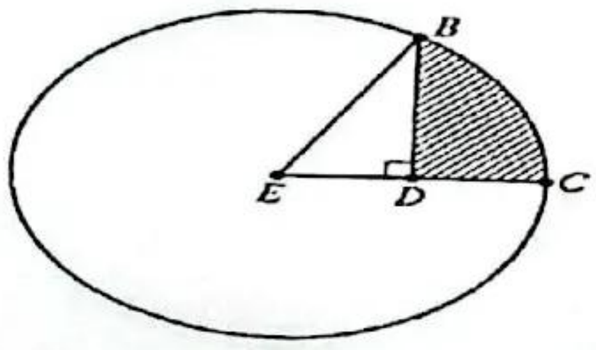
h)

$d > 0$

The total interest earned on d dollars invested for 3 months at 11 percent simple annual interest

$\frac{11}{3}\left(\frac{d}{100}\right)$ dollars

i)



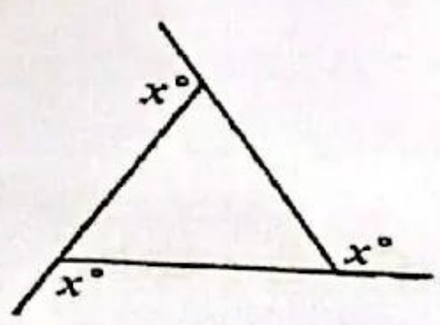
The circle with center E has radius r.

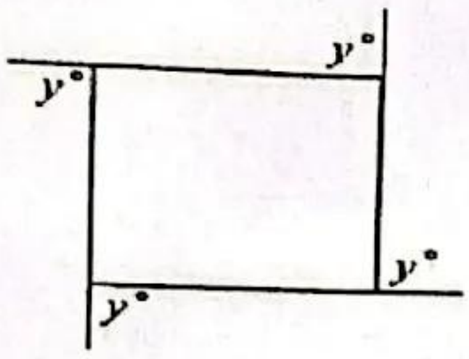
$ED = \frac{r}{2}$

The area of $\triangle EBD$

The area of the shaded region

j)

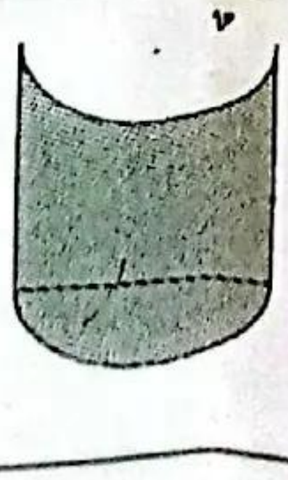




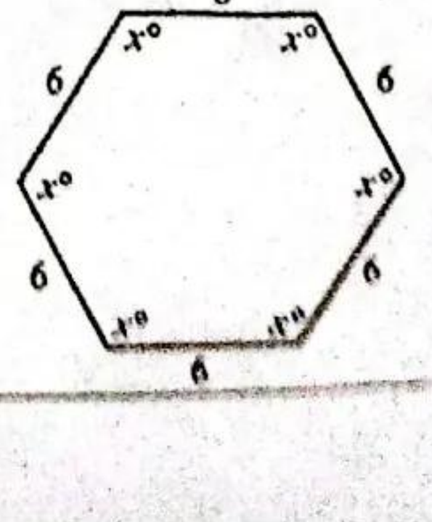
$x - y$

$\frac{x + y}{7}$

k) If each curved side in the figure above is a semicircle with radius 20, and the two parallel sides each have length 100, what is the area of the shaded region?



l) What is the area of the hexagonal region shown in the figure above?



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